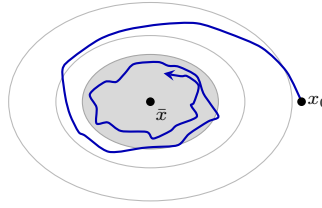


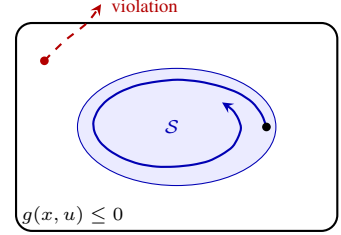
$$\Delta V(x_k) \leq -\alpha_3(\|x_k - \bar{x}\|)$$

(a) Nominal stability



$$\text{radius } \gamma(\sup_j \|w_j\|) \text{ or } \gamma(\text{tr } \Sigma)$$

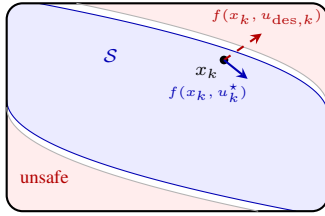
(b) Robust and stochastic stability



$$g(x, u) \leq 0$$

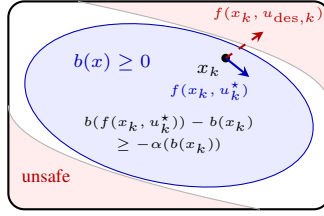
forward invariance

(c) Invariance and safety



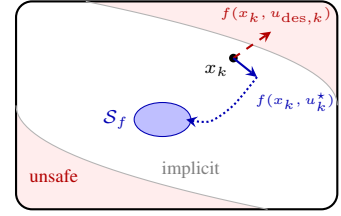
maximal safe set

(d) Hamilton-Jacobi reachability



one-step condition

(e) Control barrier function



backup trajectory

(f) Predictive safety filter

Fig. 1: Illustration of the stability and safety notions of this section. (a) Nominal stability: a Lyapunov function V decreases along closed-loop trajectories (level sets in gray) and the state converges to the equilibrium $\bar{x}$. (b) Robust and stochastic stability: under disturbances w , the state converges to a neighbourhood of $\bar{x}$ (shaded), whose size depends on the disturbance bound or the noise covariance. (c) Safety: constraint satisfaction at one time instant does not imply safety (red dashed); safety is guaranteed by an invariant set $\mathcal{S}$ contained in the constraints. (d)–(f) Safety filters: the desired input $u_{\text{des},k}$ would leave the safe set and is modified to u_k^* ; the unsafe states are shaded red. HJ reachability computes the maximal safe set $\mathcal{S}$ offline, the largest invariant set within the constraints. A CBF certifies invariance of $\{b(x) \geq 0\}$ by the one-step condition on $b(x_{k+1})$. A PSF does not compute the safe set but certifies safety by a backup trajectory (dotted) into a smaller terminal set $\mathcal{S}_f$.